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Department of Mechanical and Mechatronics Engineering

Development of a laminate calculator based on Classical Laminate Plate Theory

A report prepared for:
Waterloo Rocketry Team
Waterloo, Ontario

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Dear Professor Collins,

This report, titled “Development of a laminate strain and stress calculator based on Classical Laminate Theory”, was prepared for the Waterloo Rocketry team as my 2B Work Term Report. The Waterloo Rocketry team is a student design team that researches, designs, builds, and launches hybrid rockets for an international competition, the Spaceport America Cup (SAC), thereby providing opportunities for students to gain hands-on experience and tackle interesting problems related to rockets.

As a member of the student design team, my role is that of Airframe Lead/Team Co-Lead. My tasks include project management, team administration, participating in STEM outreach events, and design and fabrication of the structural system of the rocket.

The purpose of this report is to document the process of semi-automating the calculation of strains and stresses in composite laminates, the limitations of the calculator, and areas of future improvement. This report was written entirely by me and has not received any previous academic credit at this or any other institution.

Sincerely Yours,

A handwritten signature in black ink, appearing to read 'Shirley Kong', with a stylized, flowing script.

Shirley Kong

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Summary

The purpose of this report is to document the implementation process of a composite laminate stress and strain calculator based on Classical Laminate Plate Theory (CLPT) for use by Waterloo Rocketry. The main benefit of composite materials lie in their high strength-to-weight ratios, but due to their anisotropy they must be analyzed differently from more commonly used isotropic materials such as metal. CLPT is a model that allows the analysis of laminates to be done without resorting to Finite Element Methods (FEM) by making certain simplifying assumptions. In this report, some existing solutions to the challenge of composite material analysis are presented. Aspects of CLPT are introduced and the implementation of CLPT in code is described. It is found that the calculator that is the subject of this report lacks critical functionality, and its capabilities are too limited to be considered a final product. The limitations of the calculator are listed for further improvement, and additional recommendations of next steps are made.

1. Introduction

Waterloo Rocketry designs, manufactures, and launches a hybrid rocket each year as part of an international competition known as the Spaceport America Cup. Hybrid rockets use combustion reactants that are in different phases, most commonly liquid oxidizer and solid fuel, as is the case with the Waterloo Rocketry Team. The rocket, named Kraken of The Sky (KoTS), is 6” in diameter and approximately 15’ long and contains supporting electrical and parachute systems [1].



The airframe structure is composed of both metallic and composite components, and must be adequately robust to withstand the loads experienced before, during, and after launch. Composite materials, which tend to display exceptional strength-to-weight ratios, are used in the nose cone (found at the tip of the rocket), fins, and body tubes (which make up the main structure of the rocket). Aluminum couplers are primarily used to connect different sections of the rocket body together. Potential sources of peak loading before launch primarily come from the rocket assembly process and transportation on the way to competition. During launch, the airframe experiences bending moments and vibration from aerodynamic and inertial forces. Immediately after parachute deployment, the rocket experiences shock loading due to rapid deceleration at the instant the cord connecting the rocket and the parachute snap taut [1].

While an analysis of the list of external forces above is outside of the scope of this report, it demonstrates the need for rigorous, analysis-driven design of composite components. Due to the anisotropy of composite materials, analysis of the internal strains and stresses poses unique challenges not seen with the more widespread metallic fabrication materials which can be assumed to be isotropic [2]. The purpose of this report is to document the implementation, limitations, and areas of improvement of a simple calculator for orthotropic laminates, such that the calculator does not appear as a “black box” to future users and is open to added functionality in the future.

2. Symbol definitions

$[Q]$: Stiffness matrix for each individual ply

N_i : Force in a principal direction, where the i subscript can be x , y , or xy

M_i : Moment in a principal direction, where the i subscript can be x , y , or xy

σ_i : Stress in a principal direction, where the i subscript can be x , y , or xy

ε_i^0 : Midplane strain in a principal direction, where the i subscript can be x , y , or xy

k_i^0 : Midplane curvature in a principal direction, where the i subscript can be x , y , or xy

z : Height above or below the midplane

3. Existing Solutions

While the use of composite materials is relatively uncommon compared to other fabrication materials, a number of both free and commercially available composite analysis software exist.

3.1 Vectorlam Cirrus 3.0

Vectorlam Cirrus 3.0 is a free cloud-based proprietary software provided by Vectorply, a composite reinforcement fabric manufacturer (Figure 1). Users build their laminate by selecting parameters such as fabric, volume fraction, weight fraction, angle, and cost for each ply [3]. Using Classical Laminate Plate Theory, the software then outputs a wide variety of global laminate properties that can be exported in Comma Separated Values (CSV) format such as stiffness values and Poisson's ratio, as well as costs. It also includes a stress and deflection analysis tool for two specific geometries: beam and plate.

Product	% Fiber Wt.	% Fiber Vol.	Top Up/Dn	Rotation	Fiber Wt. oz/sq yd	Thickness in	Total Wt. lb/sq ft	Fiber Cost	Total Cost \$/sq ft
C-BX 0900	0.47619%	0.35298%	Up	0°	8.922	0.019	0.130	0 \$/lb	0.00
C-BX 0900	47.619%	35.298%	Up	0.00°	8.922	0.019	0.130	0 \$/lb	0.00
C-BX 0900	47.619%	35.298%	Up	0.00°	8.922	0.019	0.130	0 \$/lb	0.00
C-BX 0900	47.619%	35.298%	Up	0.00°	8.922	0.019	0.130	0 \$/lb	0.00
C-BX 0900	47.619%	35.298%	Up	0.00°	8.922	0.019	0.130	0 \$/lb	0.00
C-BX 0900	47.619%	35.298%	Up	0.00°	8.922	0.019	0.130	0 \$/lb	0.00

Figure 1: Laminate design page in Vectorlam Cirrus 3.0 [3]

The main drawback of Vectorlam Cirrus 3.0 lies in its lack of customizability. Fabrics are selected from a limited database of fabrics made by different manufacturers, but does not include every possible type of fabric commercially available.

3.2 Ansys Composite PrePost (ACP)

ACP is one of many Ansys design and modelling software suites, and uses the Finite Element Method (FEM) to perform stress, deflection, and failure analysis of composite materials. The workflow allows for design and analysis to occur in ACP which allows the design process to be streamlined [4].

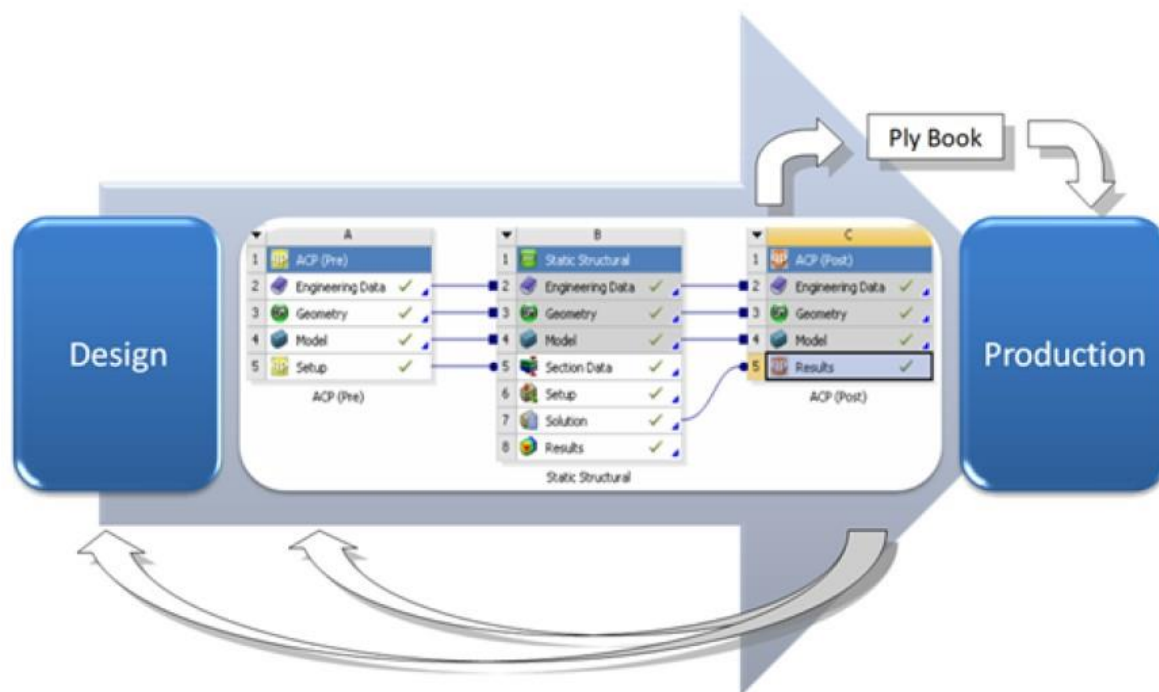


Figure 2: ACP workflow [4]

While ACP is a powerful tool when used correctly, it is much more difficult to quickly compare the effects of different laminate stacking sequences due to the heavy processor requirements when solving the model. In addition, the degree of proficiency required to obtain meaningful results means there is a risk of obtaining misleading or incorrect results, which can be harmful when taken without any means of checking the degree of accuracy. It is also relatively inaccessible due to the cost of the software.

4. Objective

The objective of the laminate calculator is to allow the user to iterate on different laminate stacking sequences and material properties to compare the effects of different laminate design changes, while allowing the user to understand the principles behind the analysis.

4.1 Criteria

The calculator must:

- Allow user inputted ply material properties
- Allow user inputted stacking sequences
- Allow user inputted ply thicknesses
- Output local ply stresses that can be used as inputs to laminate failure criteria
- Be faster than setting up and running simulations in Ansys or comparable FEM software

4.2 Constraints

The calculator should:

- Be easy to use
- Be able to read and export common file formats

5. Background

While a full explanation of CLPT is outside of the scope of this report, this section will outline the assumptions of the model, provide background on CLPT insofar as the required calculator inputs are concerned, and list the high-level steps required to arrive at the desired outputs.

5.1 Classical Laminate Plate Theory

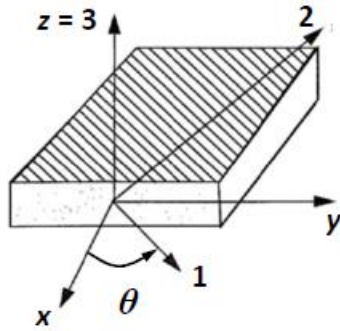


Figure 3: Diagram showing arbitrary angle between local and principal coordinate systems [2]

As shown in Figure 3, a distinction must be made between the local ply coordinate system (1,2,3) which is aligned with 1 pointing in the fiber direction, and the principal laminate coordinate system (x, y, z).

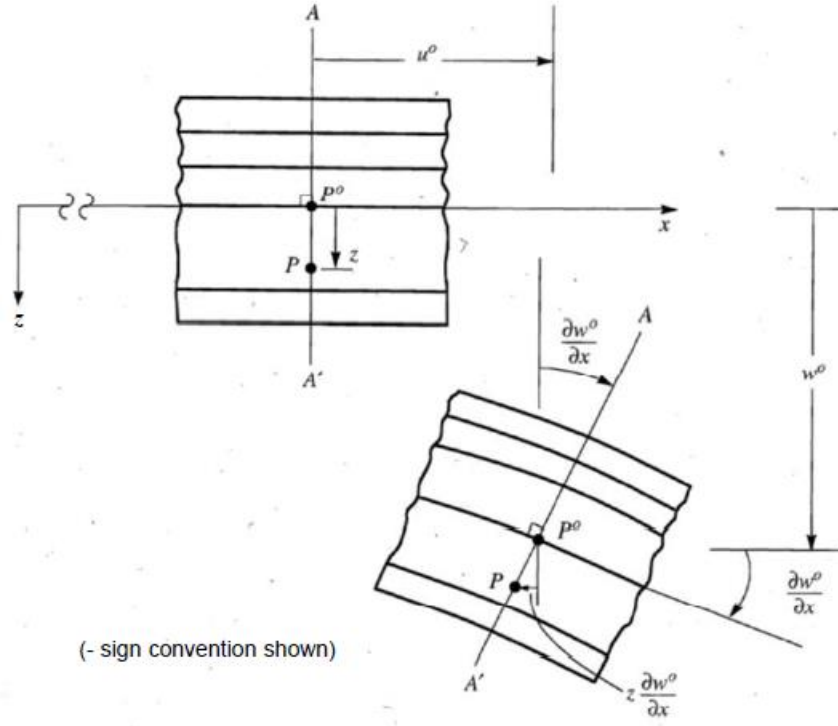


Figure 4: Diagram of cross-section of plate model used in CLPT [2]

Figure 4 shows a cross-section of the plate model used in CLPT. The plate is composed of a stack of plies of arbitrary material properties, thicknesses, and fiber orientation. When subjected to a normal force and bending moment, the laminate experiences normal strain and curvature. The laminate normal strains are expressed in Equation 1.

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + z \begin{Bmatrix} k_x^0 \\ k_y^0 \\ k_{xy}^0 \end{Bmatrix} \quad \text{Equation 1}$$

CLPT assumptions [2]:

- Plies are perfectly “bonded”
- Plies exhibit plane stress states
- All plies treated as effectively homogeneous layers

- Plies undergo linear elastic deformation
- Plate undergoes small strains and rotations
- The line perpendicular to the laminate midplane (A-A') remains straight and perpendicular after deformation
- Line A-A' does not change length

Because each ply is assumed to be in a state of plane stress, a 3x3 matrix captures the material properties of each ply. This matrix, $[Q]$, is known as the stiffness matrix and is a function of the Young's Moduli, Shear Moduli, and Poisson's Ratios of the ply. The transformation matrices defined in Equation 2 and Equation 3 are applied to $[Q]$ to transform it from local to principal coordinate systems, resulting in $[\bar{Q}]$. As shown in Equation 4, $[\bar{Q}]$ can be thought of as Young's Modulus in matrix form and can be multiplied by principal strains to calculate principal stresses [2].

$$T_1 = \begin{bmatrix} m^2 & n^2 & 2mn \\ n^2 & m^2 & -2mn \\ -mn & mn & m^2 - n^2 \end{bmatrix} \quad \text{Equation 2}$$

$$T_2 = \begin{bmatrix} m^2 & n^2 & mn \\ n^2 & m^2 & -mn \\ 2mn & 2mn & m^2 - n^2 \end{bmatrix} \quad \text{Equation 3}$$

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_{xy} \end{Bmatrix} = [T_1]^{-1}[Q][T_2] \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = [\bar{Q}] \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} \quad \text{Equation 4}$$

Where $m = \cos(\theta)$ and $n = \sin(\theta)$.

As can be seen in Equation 5 and Equation 6, forces N_i and moments M_i are functions of σ_i and z (height above or below the midplane). σ_i was previously shown to be a function of the stiffness matrix $[Q]$ and ε_i , indicating that ε_i can be calculated given $[Q]$ and the input forces and moments [2].

$$N_i = \int_{-H/2}^{H/2} \sigma_i dz \quad \text{Equation 5}$$

$$M_i = \int_{-H/2}^{H/2} \sigma_i z dz \quad \text{Equation 6}$$

This leads to the laminate constitutive equations in Equation 7, and Equation 8, where the matrices A, B, and D are collectively known as the laminate stiffness matrix.

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{Bmatrix} k_x^0 \\ k_y^0 \\ k_{xy}^0 \end{Bmatrix} \quad \text{Equation 7}$$

$$\begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{Bmatrix} k_x^0 \\ k_y^0 \\ k_{xy}^0 \end{Bmatrix} \quad \text{Equation 8}$$

4.2 Steps

In summary, the process used in CLPT for any given point in the laminate is [2]:

1. Calculate the ABD matrices

2. Determine midplane strains and curvatures ε_i^0 and k_i^0
3. Determine the global strain ε_i and stress σ_i variations through the laminate thickness at the point
4. Calculate the ply strains and stresses along the primary material directions (1,2,3) at the

6. Implementation

The following parameters are used as inputs to the calculator (Appendix A line 9), which is written in Python 3.8.3:

- Ply stiffness matrix $[Q]$ (assuming plies have identical material properties)
- Stacking sequence (angle of each ply relative to the global coordinate system)
- Individual ply thicknesses from top down
- Input forces and moments

A new array “height” of the absolute distances from the surface of each ply to the midplane is defined to aid in calculations (Appendix A line 28).

A matrix transformation function is defined to aid in the calculation of the ABD matrices, which requires the local ply stiffness matrix $[Q]$ to be transformed to align with the laminate coordinate system, becoming $Qbar$ (Appendix A line 35).

The ABD matrices are calculated based on Equation 9, Equation 10, and Equation 11 by means of three nested loops (Appendix A line 56). The innermost loop performs the integration through

the thickness of the laminate. The outer two loops allow the result of each integration to be stored in the appropriate array index locations [2].

$$A_{ij} = \sum_{k=1}^n (\bar{Q}_{ij})_k (h_k - h_{k-1}) \quad \text{Equation 9}$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^n (\bar{Q}_{ij})_k (h_k^2 - h_{k-1}^2) \quad \text{Equation 10}$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^n (\bar{Q}_{ij})_k (h_k^3 - h_{k-1}^3) \quad \text{Equation 11}$$

The individual 3x3 ABD matrices are then combined in one 6x6 matrix, inverted, and multiplied by the 6x1 “force_moment” column vector to obtain a 6x1 column vector composed of 3 strains and 3 radius of curvatures (Appendix A line 76). Equation 12 summarizes this calculation.

$$\begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \\ k_x^0 \\ k_y^0 \\ k_{xy}^0 \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix}^{-1} \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{Bmatrix} \quad \text{Equation 12}$$

In addition to global laminate strains, the calculator also outputs a graph showing strains ε_x , ε_y , and ε_{xy} throughout the thickness of the laminate (Appendix A line 79).

It is of note that the laminate calculator is capable only of steps 1-3 of the CLPT method for principal strain only. As such, it is incomplete. Recommendations for further improvement such as implementation of step 4 are discussed in section 7. Next Steps.

7. Validation

The calculator was validated with a test case with the following parameters [2]:

Stacking sequence and ply thicknesses: [0/90/90/0], 0.15mm

Stiffness matrix $[Q]$ given as:

```
[[1.557e+09 3.020e+07 0.000e+00]
 [3.020e+07 1.216e+08 0.000e+00]
 [0.000e+00 0.000e+00 4.400e+07]]
```

Input forces and moments, and resultant strains given as:

forces	moments	strain
[[5000]	[[0]	[[0.01]
[0]	[0]	[0.]
[0]]	[0]]	[0.]
		[0.]
		[0.]
		[0.]]

Inputting the given parameters results in the following A, B, and D matrices, and strains:

```

A
[[503580. 18120.    -0.]
 [ 18120. 503580.    0.]
 [   -0.    0. 26400.]]
B
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
D
[[ 0.0248  0.0005 -0.    ]
 [ 0.0005  0.0054  0.    ]
 [-0.      0.      0.0008]]

Strains (ex, ey, exy) and radii of curvature (kx, ky, kxy):
[[ 0.0099]
 [-0.0004]
 [ 0.     ]
 [ 0.     ]
 [ 0.     ]
 [ 0.     ]]

```

Because the input forces are normal forces only and the B matrix is empty, it follows that the laminate exhibits no curvature in any direction. The calculated strains are consistent with the given resultant strains, as the ε_y term is assumed to have been rounded to 0 in the test case.

Of note is the fact that the B matrix and A_{xy} , A_{yx} , D_{xy} , and D_{yx} terms are zero. This makes sense, because the test case laminate is a special type of laminate known as a balanced symmetric laminate. In balanced laminates, for every ply that is at an angle $+\theta$ other than 0 degrees or 90 degrees there is another ply at angle $-\theta$. In the case of the test case laminate, all plies lie at either 0 degrees or 90 degrees, which makes it balanced. Symmetric laminates are symmetric about the midplane, which also describes the test case laminate [2].

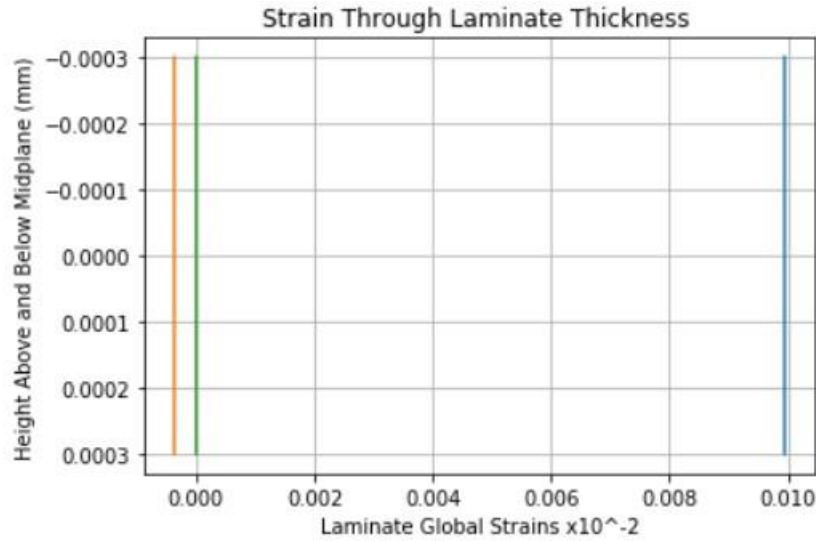


Figure 5: Strains x , y , xy through laminate thickness

Figure 5 shows the graph of ϵ_x , ϵ_y , and ϵ_{xy} through the thickness of the laminate. All strains appear as vertical lines because in the load case for this laminate, the normal forces N_x and N_y do not induce curvature due to bending, and as a result the strain does not vary through the thickness of the laminate.

8. Limitations

The laminate calculator is considered to be incomplete because it possesses significant limitations in the types of laminates that can be analyzed, and in its outputs. The calculator does not accept plies of different stiffnesses $[Q]$, or laminates with an odd number of plies. It is not yet capable of calculating stresses, and its output strains are shown in the global coordinate system. It is important to transform the outputs to local ply coordinates, because laminate failure manifests itself as individual ply failure before failure of the entire laminate. The transformed stresses can then be used in failure criteria for laminates, which are based on individual ply stresses. Lastly, the calculator has at times been found to be prone to rounding error. To avoid

this problem, the Q stiffness values are smaller than the real stiffness values by a factor of $10e2$. As a result, the calculated values must be multiplied by this factor to obtain the real results.

8.1 Next steps

In addition to addressing the limitations discussed above, the calculator is missing features that should be added to aid in the analysis of laminates. Typically, stiffness matrices are not given for a particular ply and must instead be calculated from ply material properties. The calculator should allow user inputted material properties and generate the stiffness matrix based on these inputs. More research must be done to determine what effect fiber crimping has on resultant stresses and strains [2], as woven fabrics tend to be more common for consumer use, and these effects should be considered in the calculator. Lastly, users should be able to import CSV files containing the input values, and export the calculated values to CSV files as it is a more accessible format than code.

9. Conclusion

The laminate strain and stress calculator partially attains its objectives, and there remain many improvements that must be made before it can be considered complete.

- The calculator allows user inputted ply properties, stacking sequences, and ply thicknesses. However, it requires ply properties to be constant for each ply which is impractical.
- While the calculator does output global laminate strains, it is not yet capable of calculating local ply strains and stresses.
- The calculator is faster to obtain results from than comparable composite analysis software, mainly due to its simplicity
- Currently, the calculator requires users to input their laminate parameters directly in the code. It could be much more user-friendly by incorporating a user interface for inputs or by importing CSV files.

Appendix A: Laminate Calculator Code

```
import numpy as np
import matplotlib.pyplot as plt
from math import cos
from math import sin
from math import pi

np.set_printoptions(precision=4)

#Stiffness matrix Q
Q0=[[155.7*1e7,3.02*1e7,0*1e7],[3.02*1e7,12.16*1e7,0*1e7],[0*1e7,0*1e7,4.40*1e7]]
print (np.array(Q0))
#Angle between local and global coordinate system for each ply
theta=0
```

```

#Laminate stacking sequence and thicknesses from top down
stack=[0,pi/2,pi/2,0]
thickness=[-0.15*1e-3,-0.15*1e-3,0,0.15*1e-3,0.15*1e-3]
print (thickness)
height=np.zeros(len(thickness))
midplane=int(len(thickness)/2)

#Forces N and moments M per unit length (N/m, N)
forces=np.array([5000,0,0])
moments=np.array([0,0,0])
forces.shape=[3,1]
moments.shape=[3,1]

#Create array storing the heights of k plies about the midplane
for i in range(len(thickness)):
    if (i<midplane):
        height[i]=np.sum(thickness[i:midplane])
    else:
        height[i]=np.sum(thickness[midplane:i+1])

#Transformation matrix T1 inverse and T2 (local <--> global coordinate
system)
#Function returns transformed stiffness matrix Qbar
def transform(Q0,ply_angle):
    m=cos(ply_angle)
    n=sin(ply_angle)
    T1=[[m**2,n**2,2*m*n],[n**2,m**2,-2*m*n],[-m*n,m*n,m**2-n**2]]
    T2=[[m**2,n**2,m*n],[n**2,m**2,-m*n],[-2*m*n,2*m*n,m**2-n**2]]
    T1_inv=np.linalg.inv(T1)
    Qbar=np.linalg.multi_dot([T1_inv,Q0,T2])

    return Qbar

#Calculate A, B, D matrices by integrating transformed stiffness matrix
over all plies based on laminate
#constitutive equations
#Units for A, B, D matrices respectively are hN/m, hN, hNm

A=np.zeros([3,3])
B=np.zeros([3,3])
B_inv=B
D=np.zeros([3,3])

```

```

for i in range(len(Q0)):
    for j in range(len(Q0[0])):
        for k in range(len(stack)):
            Qbar=transform(Q0,stack[k])
            ply_kA=height[k+1]-height[k]
            ply_kB=(height[k+1]**2-height[k]**2)/2
            ply_kD=(height[k+1]**3-height[k]**3)/3
            A[i,j]+=ply_kA*Qbar[i,j]
            B[i,j]+=ply_kB*Qbar[i,j]
            D[i,j]+=ply_kD*Qbar[i,j]

print("\n", "A \n",A,"\n B \n",B,"\n D \n",D,"\n")
A_inv=np.linalg.inv(A)

#Set ABD matrix, and force column vectors
ABD=np.concatenate((np.concatenate((A,B),axis=0),np.concatenate((B,D),axis=0)),axis=1)
print ("\n ABD stiffness matrix in hectanewton/meter, hectanewton, and hectanewton meter respectively: \n",ABD,"\n")
force_moment=np.concatenate((forces,moments))

#Calculate global laminate strains and radii of curvature at midplane
midstrain=np.dot(np.linalg.inv(ABD),force_moment)
print ("\n Strains (ex, ey, exy) and radii of curvature (kx, ky, kxy): \n",midstrain)

#Calculate strain in each ply k and plot
estrain=midstrain[:3]
kstrain=midstrain[3:6]
estrain.shape=[3,1]
kstrain.shape=[3,1]

plystrain=np.zeros(shape=(3,len(height)))
plystrain1=np.zeros(shape=(3,len(height)))

#Sanity check to ensure array dimensions are correct
print ("plystrain shape",plystrain.shape)
print ("height shape",height.shape)
print ("estrain shape",estrain.shape)
print ("kstarin shape",kstrain.shape)

for i in range(3):
    for j in range(len(height)):

```

```

        plystrain[i,j]=estrain[i,0]+np.dot(height[j],kstrain[i,0])
print("\n Strain for each ply k: \n",plystrain,"\n")

for i in range(3):
    plt.plot(plystrain[i,:5],height)
    plt.gca().invert_yaxis()
    plt.xlabel("Laminate Global Strains x10^-2")
    plt.ylabel("Height Above and Below Midplane (mm)")
    plt.grid(axis="both")
    plt.title("Strain Through Laminate Thickness ")

print ("forces \n",forces,"\n moments \n",moments,"\n strain \n",midstrain)

```

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